

Midterm Exercise 1: Schelling Modified

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While the original Schelling model assigns the same threshold of alike neighbors for happiness to all agents, this does not reflect the reality of individuals' housing preferences. By allowing for variability in agent preferences, we can investigate whether the distribution of agents will be dependent on their preferences for degree of similarity of neighbors. Additionally, we can understand how the minimum and maximum degree of preference influence whether segregation emerges. I seek to understand how individual preference for neighbor makeup influence where within a segregated neighborhood agents will reside as well as what maximum threshold for alike neighbors is necessary to produce segregation when the minimum is 0. I hypothesize that when segregated neighborhoods emerge, individuals with lower thresholds for happiness will be on the exterior because they are willing to live adjacent to more agents of the opposite type, whereas those with higher thresholds will move to be in interior blocks of the same type. Following this, I hypothesize that even with a minimum of 0, a relatively low maximum threshold of happiness will still yield segregation.

To investigate these questions, I have changed how the happiness threshold operates in the model. Instead of setting a global preference for the proportion of neighbors sharing the same type, in my modified model a minimum and maximum value is specified. Then upon initialization, agents are each independently assigned a value for this threshold between the specified minimum and maximum according to a uniform random distribution. This value remains constant for each agent throughout the simulation. These differences are visualized by differing opacity such that agents with higher shared alike values are more opaque. As discussed previously, this is more realistic because it allows for variability in agents. In addition, it allows us to potentially replicate the previous finding that segregation emerges with a relatively low preference for the proportion of alike neighbors, but in an even more realistic and seemingly less likely scenario where some agents have almost no preference.

Though I have not completed any batch runs, I now consider select end-states of the model with parameter values of interest. In each of the following runs, I hold the following at the same constant values specified below and vary only Desired Share Alike Min and Desired Share Alike Max.

- Random Seed: 42
- Width: 30
- Height: 30
- Population Density: 0.7
- Share Type 1 Agents: 0.7
- Vision Radius: 1

In figure 1 below, I consider the maximal distribution of agent preferences, with desired share alike values between 0 and 1. Similar to the original Schelling model, this does not resolve as some agents are unable to find suitable locations. Notably, unlike the original model, we do have clear segregation occurring. As expected, however, it is the agents with extremely high desired share alike thresholds who are unable to find a location where they are happy and thus continue to move each turn.

In figure 2, I consider a range of thresholds between 0 and 0.9. Now we do see a stable end state which, expectedly, is very segregated. At first glance, it's not immediately obvious whether desired share alike and location are connected. It does appear that agents with high thresholds for happiness exist on the exteriors of neighborhoods where there are open spaces serving as boundaries and buffers between neighborhoods. On the other hand, boundaries in which neighborhoods are directly contiguous are occupied by those with lower thresholds.

Finally in figure 3, I consider a range of thresholds between 0 and 0.5. Even with this relatively high maximum desired share alike value, the degree of segregation to my eye looks very small. There are some small pockets of the same agent type, but the large homogenous neighborhoods no longer exist. The clustering of similar agents is much more local and small-scale.

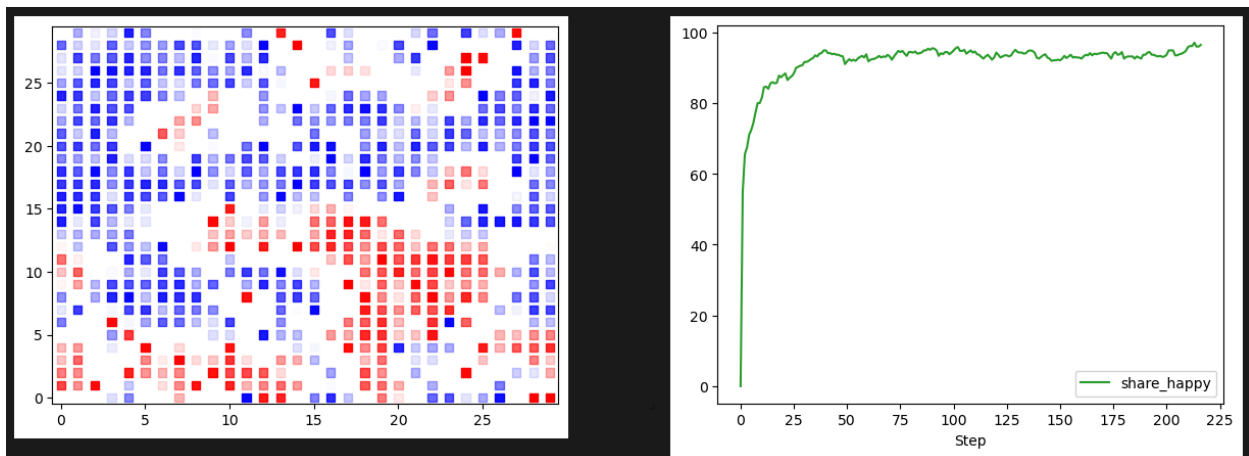


Figure 1: With a range of 0 to 1 for desired share alike, the model does not reach a stable state.

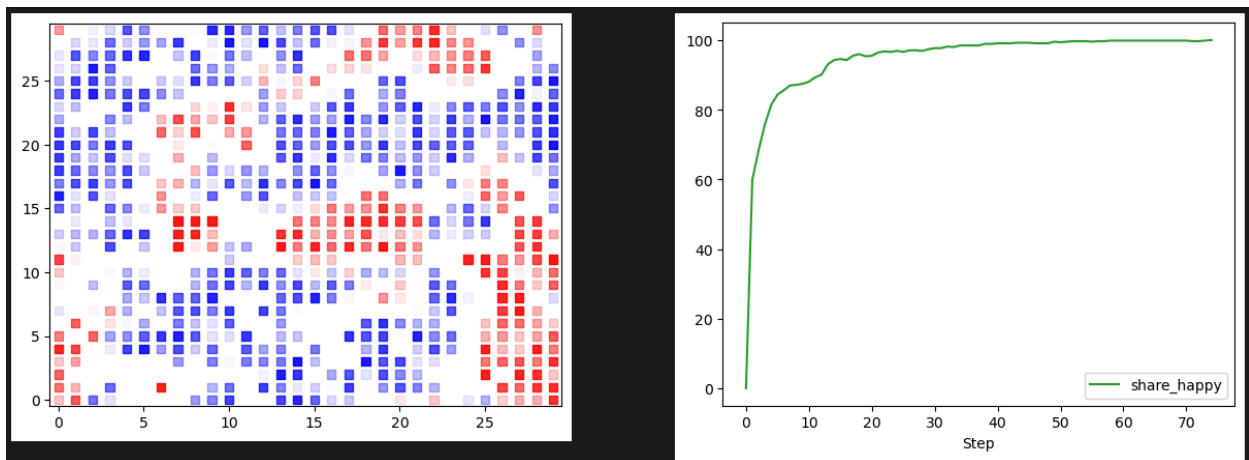


Figure 2: With a range of 0 to 0.9 for desired share alike, the model reaches a stable state with high segregation.

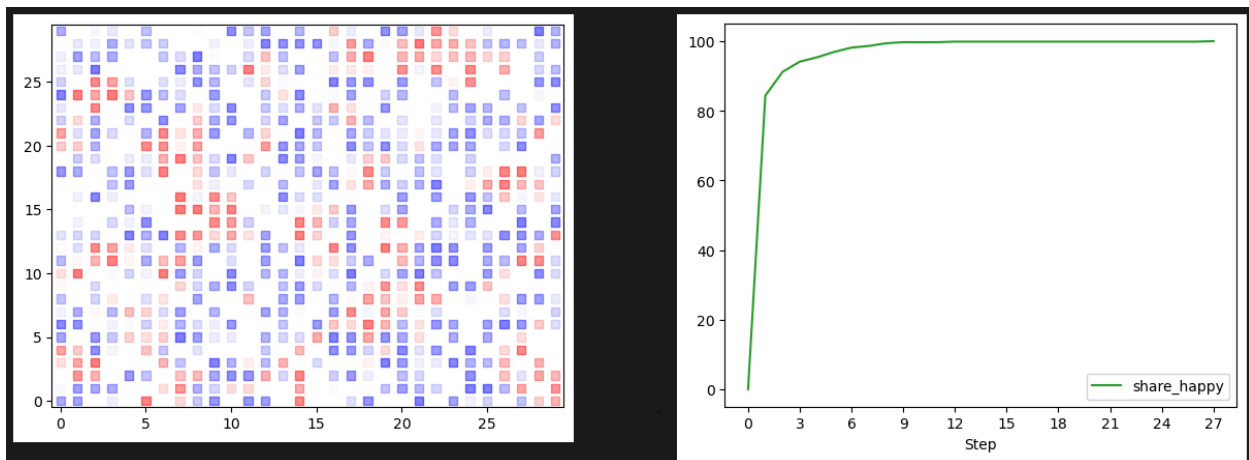


Figure 3: With a range of 0 to 0.5 for desired share alike, the model reaches a stable state with only minimal, small-scale neighborhoods of the same agent type.

Unexpectedly, there does not seem to be an difference between interior and exterior location of agents. The difference occurs along the boundaries of neighborhoods. High degrees of segregation also do not occur until relatively high maximum desired share alike values exist, a sharp contrast to the original Schelling model.

Code GitHub Link:

https://github.com/gabereichman/ABM_Assignments/tree/main/schelling_modified

Resource Usage Statement:

Due to the relatively minor coding requirements to implement this change, I just used the MESA documentation to see what AgentPortrayalStyle allowed/expected for its color parameter input.